

SymPy Bug Card

Bug 27: solveset returns a false acosh solution outside the principal range

Task. Solve the principal inverse hyperbolic equation over the complex numbers.

$$\operatorname{acosh}(x) = 2\pi i$$

SymPy's answer.

$$\operatorname{solveset}(\operatorname{acosh}(x) = 2\pi i, x, \mathbb{C}) = \{1\}.$$

Correct answer.

$$\emptyset$$

Explanation. Substituting SymPy's returned value gives $\operatorname{acosh}(1) = 0$, so the residual is $-2\pi i$. The mistake is that $\cosh(2\pi i) = 1$ solves only the composed equation after applying $\cosh$, not the original principal-branch equation.

Likely root cause. In `sympy/solvers/solveset.py`, the generic complex inversion path maps targets through $\operatorname{acosh}^{-1} = \cosh$ without checking the principal range of acosh . This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-027-solveset-returns-a-false-acosh-solution-outside-the-principa/](#)